ASAP7 Predictive Design Kit Development and Cell Design Technology Co-optimization

Invited Paper

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***Abstract*— This work discusses the ASAP7 predictive process design kit (PDK) and associated standard cell library. The necessity for multi-patterning (MP) techniques at advanced nodes results in the standard cell and SRAM architecture becoming entangled with design rules, mandating design-technology co-optimization (DTCO). This paper discusses the DTCO process involving standard cell physical design. An assumption of extreme ultraviolet (EUV) lithography availability in the PDK allows bi-directional M1 geometries that are difficult with MP. Routing and power distribution schemes for self-aligned quadruple patterning (SAQP) friendly, high density standard cell based blocks are shown. Restrictive design rules are required and supported by the automated place and route (APR) setup. Supporting sub-20 nm dimensions with academic tool licenses is described. The APR (QRC techfile) extraction shows high correlation with the Calibre extraction deck. Finally, use of the PDK for academic coursework and research is discussed.**

Keywords*— *Standard Cell Library; Automatic place and route; Timing Characterization; Design Rules, EUV; finFET; Parasitic Extraction; Physical Design.

I. INTRODUCTION

Recent years have seen the predominance of finFETs to alleviate short channel effects, improve leakage, and enable some continued V_{DD} scaling. However, a state of the art finFET-based PDK for modern circuit and physical design academic courses and research has been lacking. The finFET-based FreePDK15 [1] included a standard cell library, but lacks full layout vs. schematic (LVS) verification and parasitic extraction. Consequently, the only fully capable sub-65 nm educational PDKs are the planar CMOS based Synopsys 32/28 nm and the 45 nm FreePDK45 [2][3]. These libraries use very large cell heights, in contrast to recent industry trends. Moreover, they either lack SRAM cells and special design rules for them, or have unrealistic layouts provided.

Commercial libraries and PDKs, especially for advanced nodes, are often difficult to obtain for academic use, and access to the actual physical layouts is even more restricted. Furthermore, the necessary non-disclosure agreements (NDAs) are un-manageable for large university classes and the plethora of design rules can distract from the key points. NDAs also make it difficult for the publication of physical design as these may disclose proprietary design rules and structures.

This paper describes the ASAP7 PDK development, focusing on design/technology co-optimization for SRAM and standard cells. The focus is on the tradeoffs and development of a cell library publicly available for academic use. The PDK contains all the views and verification capabilities for custom or automated placed and routed designs, as well as the ability to mix both.

A. Contribution

The cell library in this paper was developed at Arizona State University using the ASAP7 7 nm finFET-based predictive PDK [4-7] that allows realistic physical design at advanced nodes. It supports full physical verification (DRC and LVS) and parasitic extraction through Mentor Graphics Calibre. The standard cell library enables auto-placed and routed designs. The impact of lithography assumptions are discussed in this context. The library described includes cells in four threshold voltages, allowing high performance for processors to very low leakage for internet of things (IOT). Non-state of the art (academically licensed) APR tool usage with sub-10 nm design rules are described, and good parasitic correlation is shown. Some example APR designs using the library are briefly described.

B. Paper Organization

Section I has provided the paper background. Section II comprises a PDK overview, including the various front-end-of-line (FEOL), middle-of-line (MOL) and back-end-of-line (BEOL) layers. Cell library aspects and electrical performance are described in Section III. Section IV describes the scaling of the technology library exchange format (techLEF) file, cell LEFs, and the QRC techfile to allow APR with sub-20 nm features using the academically available Innovus licenses. Good correlation between Calibre parasitic extracted nets and those estimated using the Innovus QRC techfile is demonstrated. APR experiments on small (4k gate) designs as well as large (350k) gate designs are described in Section V, along with the impact of different V_{th} cell libraries. Section VI concludes the paper.

II. PDK OVERVIEW

A. Layer and Physical Design Overview

The basic 7.5T cell layout shown in Fig. 1 contains the key structures of the FEOL through M1. Liebmann, *et al*. assume a

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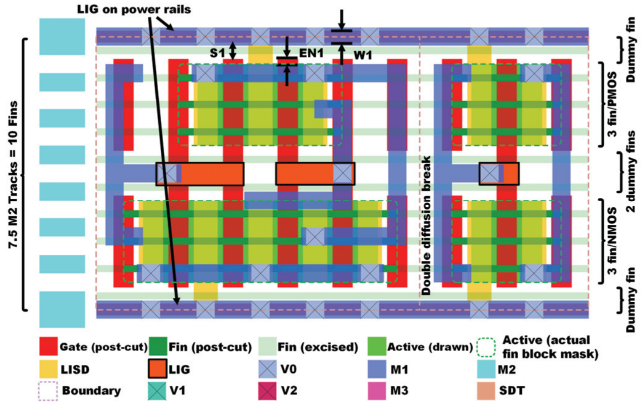

Fig. 1. A 7.5 M2 track standard cell template, with 3-fins per device type, shows the FEOL, MOL, and M1 in adjacent NAND2 and inverter. S1 = minimum LIG to GATE spacing, EN1 = minimum GATE endcap, W1 = minimum LIG width.

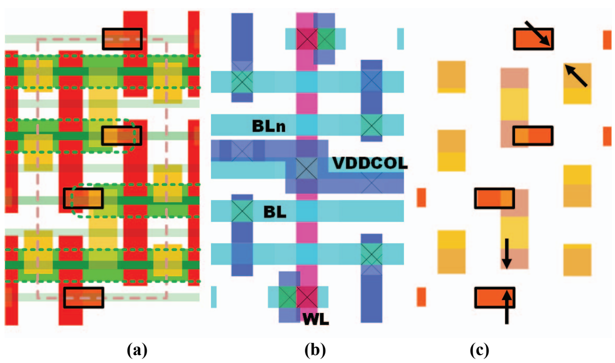

Fig. 2. 111 SRAM cell with M2 BLs matching the standard cells (a) and (b). This cell has good FEOL lithography. The SDT to LIG limitations (c).Since these layers are EUV, rounding will actually help alleviate TDDB.

24 nm N7 fin pitch [8], but ASAP7 assumes a more relaxed 27 nm fin pitch, using self-aligned quadruple patterning (SAQP) with optical immersion lithography (193i). M1 through M3 are assumed to be patterned using extreme ultra-violet lithography (EUVL) at a 36 nm pitch (18 nm width and spacing), obtained using a single exposure [9]. Unidirectional (1-D) lines can be patterned at 32 nm pitches and bidirectional (2-D) patterns have been demonstrated at sub-18 nm [10][11]. However, the PDK assumes a conservative 2-D pitch of 36 nm due to continued difficulty in moving EUV into production [12].

The contacted poly pitch (CPP) is 54 nm, again less aggressive than some publications suggest [8][13]. Gates are patterned via self-aligned double patterning (SADP) on pitch. L_{gate} is 21 nm, to reduce short channel effects and I_{off}. Raised source drains (S/Ds) are grown on the fins between the spacers, where the deposition is constrained by the gate spacers for uniformity. The self-aligned source drain trench (SDT) layer fills the opening between the spacers, connecting the SD to the MOL local interconnect (LISD) layer. MOL is assumed to be patterned using EUV. The 2nd (LIG) MOL layer connects to gates and its top is coplanar with LISD.

Back end of line (BEOL) layers M1 through M3, as well as via layers V0 through V2 are EUV patterned. It is a small change to support SAQP M2 and M3. 193i SADP is assumed for M4, M5 and M6, M7, which have pitch values of 48 nm and 64 nm. Calibre DRC decks support metal mask decomposition into colors for SADP. The M4-M7 vias are patterned with 193i LELE MP. M8-M9 and V7-V8 vias have dimensions exceeding 40 nm and can thus be patterned using 193i single exposure lithography.

MOL and BEOL design rules are largely based on time dependent dielectric breakdown (TDDB) determination after considering misalignment. All vias are self-aligned to the metal hard-mask.

B. Electrical Performance

The PDK uses BSIM-CMG SPICE models and the value used are derived from publically available sources with appropriate assumptions. Drive current increase from N14 to our predictive N7 is 15% with near ideal subthreshold slope of less than 65 mV/decade at 25ºC. PMOS strain seems to be easier to obtain according to 16 nm and 14 nm foundry data and larger PMOS than NMOS I_{dsat} have been reported [14]. Following this trend, the PDK assumes a PMOS to NMOS drive ratio of 0.9:1, allowing an inverter fan-out of six (FO6), instead of the traditional four.

The PDK supports four threshold voltages, viz. SLVT, LVT, RVT, and SRAM, to allow both high performance and low power designs. The drive strength reduces from SLVT to SRAM, with the latter support internet of things (IOT) class low-standby power designs. Typical-typical (TT) models, fast-fast (FF) and slow-slow (SS) models are also provided for multi-corner APR optimization. The TT corner per fin RVT NMOS I_{dsat} is 37.85 µA (effective current, I_{eff}, is 18.13 µA) at V_{DD} = 700 mV at 25ºC.

C. SRAM Cells

As mentioned, SRAM cells were critical to the DTCO effort in defining the ASAP7 design rules. The smallest 111 6T SRAM cell is shown in Fig. 2. While M1 bit lines (BLs) allow easier lithography, this cell is more compatible with the M2 BL direction the same as fin direction, matching the 7.5T library. The FEOL and BEOL views comprise Fig. 2(a) and (b) respectively. As in commercial processes, the SRAM well and active design rules allow tighter spacing, as well as a number of MOL rules, using an identification layer. Specifically, the SDT is not coincident with the LISD to maximize contact area without pushing SDT to LIG TDDB (Fig. 2(c)). This cell also requires write assist for good reliability [15]. The 111 and 112/122 SRAM cells are 0.023 and 0.029 μm^2 in area, roughly in line with recent foundry data. The SDT and LISD are limited by TDDB to LIG so they do not fully cover the S/D regions. This DTCO consideration required separate drawn, rather than derived SDT layers.

III. CELL LIBRARY ARCHITECTURE

A. Gear Ratio and Cell Height

Similar to planar CMOS, the selection of standard cell height in finFET process is also application specific—whether aiming at low standby power or high performance. It relates to the number of usable fins per transistor, i.e. drive strength, and the corresponding number of horizontal (here M2) tracks that can access cell pins, i.e. gear ratio, which should be an integer

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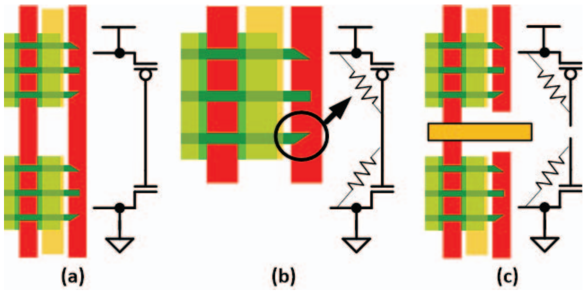

Fig. 3. Dummy gate at DDB (a). Sharp fin edges arise due to mask rounding when cutting fins. This creates a TDDB scenario between the fins and dummy gate (b). This is avoided where possible by cutting the dummy gates (c).

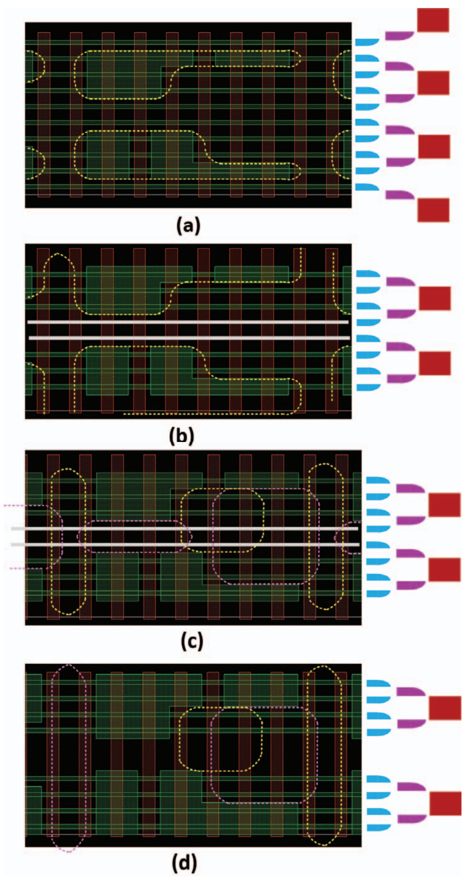

Fig. 4. The original EUV fin cut mask assumptions and SAQP spacer/mandrels (a). Another option that allows larger patterns (b). LELE fin cut (c) may be problematic, but an even number of fins eases this (d).

for fins. The 3/4 M2 to fin pitch ratio in ASAP7 allows designing standard cells at 6, 7.5, 9 or 12 M2 tracks and with 8, 10, 12, or 16 fins in the cell height, respectively. EUVL enables a compromise 7.5T library that is the focus here. Multiple foundries have mentioned "fin depopulation" on finFET processes—using fewer fins each generation. Shorter cells are enabled by diminishing need for higher speed as designs are increasingly power constrained, but are also impacted by high drive per fin and MP lithography considerations. The near 1:1 P:N drive ratio contributes as well. The minimum transistor separation dictates that the outer two and middle two standard cell fin locations may not be used, as apparent in Fig. 1. Thus, in the 7.5T cells each transistor can be one to three fins. Notwithstanding the number of fins, the transistors have sufficient drive capability due to the large per-fin drive current value (see Section II.B). ASAP7 library cells are classified based on their drive strengths, so that those with three fin transistors are said to have a 1×, while less have fractional, e.g., one fin is p33× (0.33×) drive.

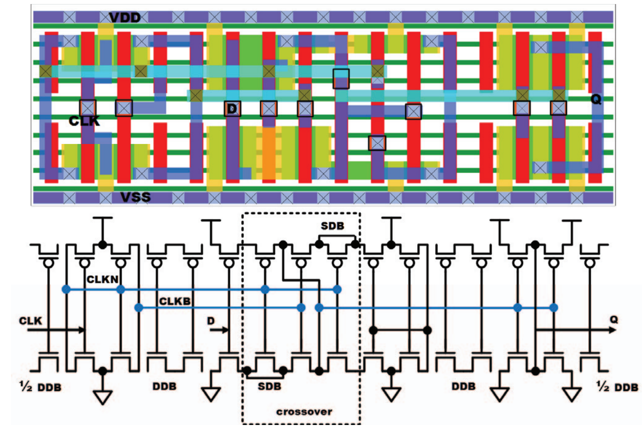

Fig. 5. Cell layout of a transparent high D-latch. Double and single diffusion breaks are shown. The latter require LISD crossovers. Fins shown are pre-cut.

B. Fin Cut Implications

Vertical extent of the drawn active layer in the PDK denotes the raised source-drain region. However, as evident in Fig. 1, the actual active layer (fin keep) mask extends horizontally halfway underneath the dummy gates. This necessitates a double diffusion break (DDB) between diffusions at disparate voltages. Therefore, all standard cells end in a single diffusion break, producing a DDB when abutted. A single diffusion break (SDB) is allowed, but design rules verify that the diffusions are shorted. SDBs are useful, particularly in the latch and flip-flop CMOS pass-gates.

As shown in Fig. 3, mask rounding effects, when implementing fin cuts, create sharp fin edges that have high charge density and consequently, a high electric field. This creates a potential TDDB issue between the fin and dummy gate, causing leakage through the gate oxide. To mitigate this, where possible, the dummy gates are cut to preclude an electrical path between the PMOS and NMOS transistors through the dummy gate. This dummy gate cut also allows better LIG routing within the cells.

SAQP requires fixed width and greatly limits the spacing of fins (Fig. 4). Assuming an EUVL fin cut/keep mask enables the 7.5T library as designed (Fig. 4(a)). If MP is assumed, LELE cut/keep is complicated as in Fig. 4(b) and (c), where the middle fins to be blocked are shown in grey. An even number of fins for P and N transistors allows the SAQP mandrel to completely define the fin locations, greatly easing the subsequent fin masking (Fig. 4(d)). A 6 or 6.5T (or 9T) library, which has an even number of fins per device at two or four, is more amenable to SAQP fin patterning.

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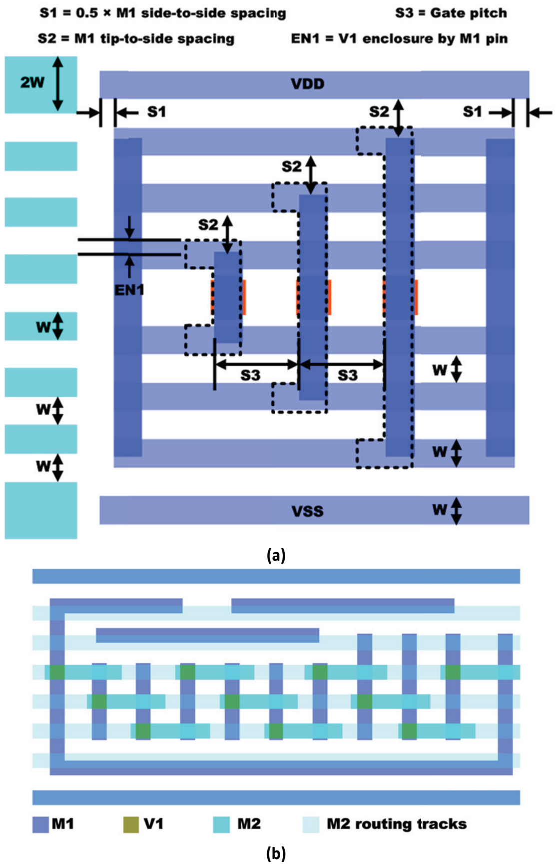

Fig. 6. M1 template to ensure and maximize pin access (a) An AOI333 cell with all of its input M1 pins connected to M2 (b).

C. Standard Cell MOL Usage

LISD is primarily used to connect diffusion regions to power rails and other diffusions within the cell, relieving M1 routing congestion, as well as to M1. Connections to the M1 power rails are completed using LIG, since using LISD instead would violate the LISD tip-to-line spacing depending on the abutting cell LISD. The PDK design rules require a 4 nm gate endcap past the active (EN1) and a 14 nm LIG to gate spacing, restricting the LIG power rail to its 16 nm minimum width. At this width V0 is not fully landed, but V0 is fully populated, ensuring robustness. The mask misalignment assumptions and their impact is described in [4]. The 2-D EUV assumption allows LISD to serve a secondary role of a routing layer within a cell, used sparingly, due to higher resistivity and larger design rules [13]. It is helpful in complex cells—especially sequential cells—due to M1 routing congestion and to limit M2 track usage. Fig. 5 shows a D-latch where LISD connects diffusions across SDBs, as well as under the M2 track. SDBs arise due to the inability to accommodate two gate contacts with an M1 crossover. They are extensively used in the CMOS pass-gate crossover structure comprised of a CLKN-CLKB-CLKN gate combination outlined in the schematic at the layout pitch.

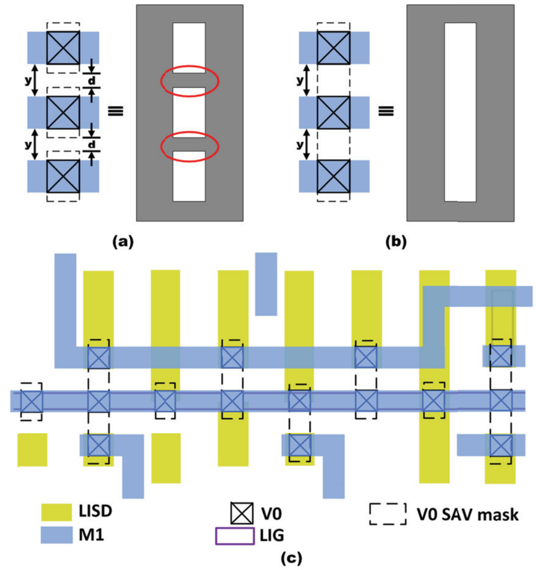

Fig. 7. (a) Sub-lithographic features (red ellipses) on SAV mask. (b) Merging SAV masks precludes such features. This is very helpful throughout the cell library, allowing on grid vias from M1 to V0 and M2 to M1 during APR (c).

D. BEOL and Via Merging

Referring to Fig. 6(a), the 7.5T standard cell results in equidistant M2 routing tracks that are all 18 nm wide, except for the 36 nm wide M2 power rails. The template accommodates eight horizontal M1 constructs that are arranged as two groups of four equidistant tracks, with the groups themselves having a larger spacing with regards to the standard cell center. Accounting for the extra M1 spacing near the center, instead of the power rails, helps to maximize pin access through M1 pin extension past the M2 tracks adjacent to the power rails. The vertical M1 pins are placed at the gate pitch. Purely vertical M1 pins do not provide the required V1 enclosure for vias from the outermost M2 tracks. Therefore, C-shaped pins (dotted polygons) are used to allow full via landing when possible and also honor tip-to-line DRC used to avoid mask error enhancement factor (MEEF) issues.

In the library, all M1 pins span at least two M2 routing tracks. This ensures that even cells with a large number of pins, such as the AOI333 in Fig. 6(b), have all pins accessible. The use of staggered M2 (by the APR) allows escapes through M3. The M1 template enables rapid cell library development, since the pre-delineated allowed shapes comprehend the design rules. Wide M2 power follow rails are allowed by the 7.5T height, but assume non-self-aligned V1, M1 pins near the power rails is enabled by via merging, which is comprehended by the DRCs. The presence of V0 on the power rails produces the cases in Fig. 7, where the actual un-merged via mask spacing d may be smaller than the minimum metal side-to-side spacing labeled y. This would result in sub-lithographic via mask features—avoiding which entails via mask spacing limited metal placement. Merging the vias permits the patterning of V0 containing M1 tracks at minimum spacing.

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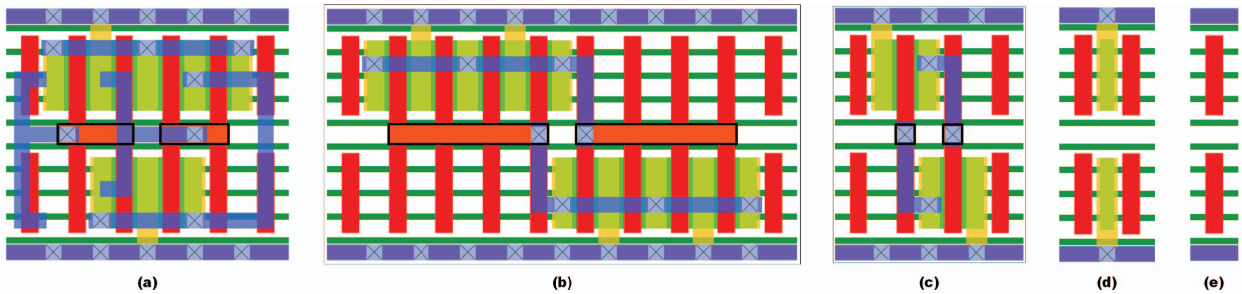

Fig. 8. (a) NOR2x1 (b) DECAPx4 (c) DECAPx1 (d) Tapcell (e) FILLER (minimum size).

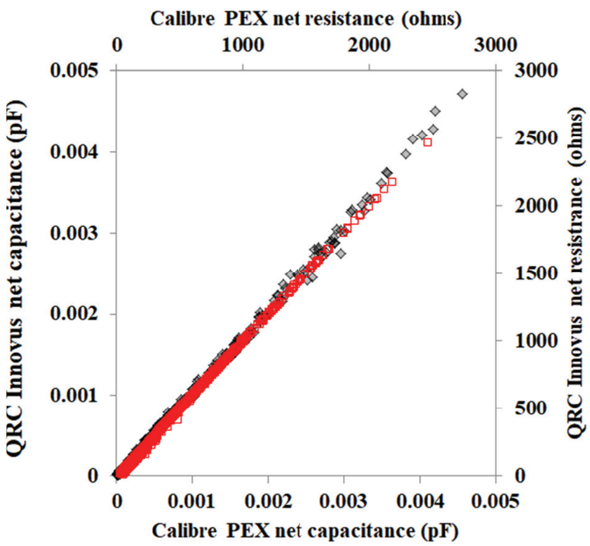

Fig. 9. Scaled QRC techfile estimated vs. actual (unscaled) Calibre PEX total net capacitance (black) and resistance (red).

IV. CELL LIBRARY DETAILS

A. Library Components

The necessary collateral is provided with the library. Physical pin and blockage views (in the .lef) are scaled by 4× as explained in Section V. Sample GDSII and Open Access schematic/symbol views for some representative cells are provided to allow users to add their own cells fitting the architecture. LVS netlists (.cdl) are provided so that extracted post-APR designs can go through LVS, and parasitic extracted (PEX) cell netlists are included for circuit simulation. Liberty files are provided using Cadence standard ECSM lookup tables. The tables are all 7×7, centered on FO6 capacitance and the resulting input slew rate. We chose FO6 since the input capacitance for the same drive current inverter is reduced in a logical effort analysis, making that appropriate for the nearly even PMOS to NMOS drive strengths. Timing characterization used Liberate, which also produces tables for leakage, active power, and setup and hold times. The liberty files are based on the PDK supplied process corners, i.e., SS, TT, and FF. V_{DD} voltages used are 630 mV, 700 mV, and 770 mV, for those corners, respectively. Separate libraries provide the four supported V_{th}'s. One V_{th} is used per cell.

136 cells are provided for each of the four V_{th}'s, including filler, well tap, tie high and tie low, and decoupling capacitance cells. There are 106 combinational logic cells, 17 scan and non-scan flip-flops, six latches, and three integrated clock gaters, encompassing multiple drive strengths. Inverters have strengths up to 13× and buffers up to 24×. And-or-invert (AOI) and compatriots, i.e., OA, AOI and OAI that do not lay out efficiently are not included. We define inefficient layout by excessive DDBs or where the cell can be assembled in a more area efficient manner by the APR tool using other library cells. Our reasoning is that nothing is gained by gluing two gates together as a cell, but placement flexibility is sacrificed. Representative cells are shown in Figs. 1 and 8. Each cell ends at the left and right side with 1/2 of a DDB. NAND2 abutting INVx1, and NOR2x1 with three fins per device comprise Figs. 1 and 8(a), respectively. In general, we optimized drive to area rather than attempting to make balanced rise and fall times. This results in the most area efficient library from a drive vs. area perspective but requires care in choosing cells for clock tree synthesis.

Decoupling capacitance is generally inserted opportunistically as fill in real APR designs and thus DECAP cells are provided in the library. Our usage generally fills with DECAP from wide 10 poly track (Fig. 8(b)) progressing to smaller sizes to maximize capacitance. Fig. 8(c) shows the four poly track decap cell. We insert extra tap cells (Fig. 8(d)) for two track fill and finally the single poly track fill cell shown in Fig. 8(e). At present antenna rules and diode cells are not supported in the PDK or cell library.

B. Cell Performance

The RVT TT corner gate delays including extracted parasitics for the 1x inverter (INVx1) driving a FO6 load is 5.7 ps and 7.1 ps for falling and rising edges, respectively. NAND2x1 delays are 5.9 ps and 9.5 ps, while the NOR2x1 has delays of 7.7 ps and 7.3 ps. The SLVT (super-low V_{th}) inverter provides rising and falling delays of 5.1 ps and 4.6 ps, respectively, while the SRAM (highest V_{th}) INVx1 has 10.7 ps and 9.1 ps delays at the nominal voltage and typical corner. The SRAM V_{th} I_{off} is approximately three orders of magnitude lower than that of the SLVT. The other V_{th}'s, i.e., RVT and LVT, are approximately in between with an order of magnitude I_{off} difference between them. SS, TT and FF corners are supported in the library.

V. APR USAGE

A. Scaled LEF and QRC TechFile

The library validation used Cadence Innovus as the APR tool. Academic licensing does not allow features below 20 nm. To alleviate this issue, some of the APR collateral is scaled for

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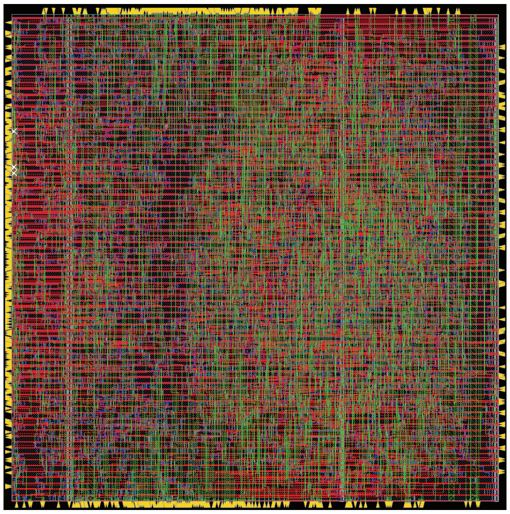
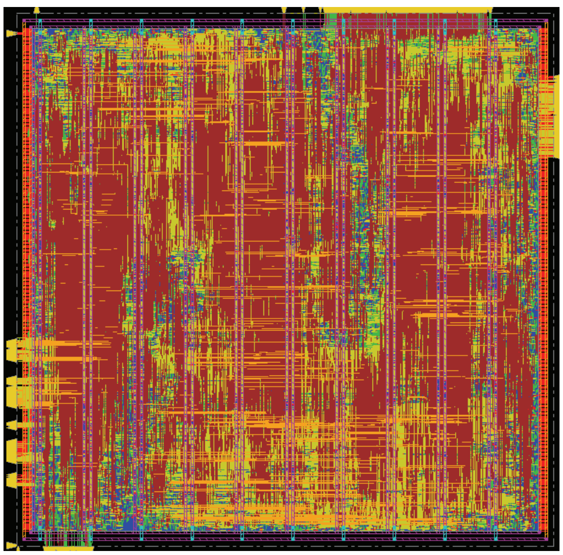


(a)



(b)

Fig. 10. APR designs using the 7.5T cell library. A small (4k gate) level-2 cache EDAC routed only on 1-D M2 and M3 (a) is 22×22 µm. A large AES engine for spacecraft that is 250 µm on a side using M2 to M6 for routing (b).

APR, with the resulting design .gds shrunk when written out. The technology LEF and macro LEF are scaled up by a factor of four. After completing the APR, the exported GDSII is imported a stream-in scale factor of 0.25, to obtain a PDK compliant design.

To compensate the parasitic extraction in Innovus to the scaled LEF geometries, the interconnect resistivity and the dielectric permittivities are adjusted in a scaled QRC techfile. We ran APR on small designs and constrained the metal layers to provide high wire density. The QRC based SPEF was then compared to the SPEF obtained from Calibre PEX after importing the design into Virtuoso. Fig. 9 shows the resistance and capacitance correlation obtained. The Calibre PEX and the scaled QRC technology file show a 97.8% correlation for total net capacitance, and a 99.1% correlation for total net resistance. We believe this is adequate to make the library usable, since the error is less than on-die random variability. Ideally, Cadence will provide sub-20 nm licenses in the standard academic licensing soon. We note that Synopsys ICC has the same academic licensing issues [5].

B. *Design Experiments*

Using the techLEF, we use cell height variable track patterns to build M2 special routes over existing M1 follow rails. This provides the correct 7.5 M2 pitch at the cell level, while providing minimum M2 spacing to match the over cell routing M2 grid shown in Fig. 5. Modifications to align the cell library with an M2 and M3 SAQP assumption requiring equal spaces between metals is easily supported, although it will require non SAV V1 on the power follow rails. M2 follow rail width is up to the APR scripts, i.e, the designer. This is not currently in the DRCs.

The technology LEF has fixed via definitions and via generation rules definitions for all layers. As mentioned, they are assumed SAV, allowing upper metal width vias. Quality multi-cut via generation for wide power stripes and rings, essential to low resistance power delivery, may require user modification for different sizes than we have set up. Single cut vias are used for signal routing.

C. *Results*

We primarily used two designs to validate the cell library in an APR flow. We use these designs as we are familiar with them and the large one has been implemented on 90 nm silicon. The first, a cache EDAC pipeline stage is small and allows fast experiments, which implements Hamming code based 2-bit error detection and 1-bit encoding on the cache input direction and syndrome based correction in the output direction. This design has 535 top-level input and output pins and about 4k instances, varying with APR optimization. This design works in single clock domain with its top metal routing layer restricted to M3. We have validated use of single V_{th} flows (with one of RVT, LVT, SLVT, SRAM) and mixed V_{th} flows, as well as multi-corner optimization. The resulting layout comprises Fig. 10(a) and have achieved cell area utilization above 90%. The leakage follows the cell library V_{th} chosen, following the PDK I_{off} values. The example here assumes one dimensional SAQP compliant routing for M2 and M3. The mixed V_{th} implementation allows better than 5 GHz clock rate. Innovus timing optimization with useful skew on the clocks allows 6 GHz at the TT 25ºC corner, where SLVT dominates the cell usage.

The second design is a triple modular redundant advanced encryption standard (AES) engine with fully unrolled 14 stage pipelines. We used this design as it is large, but requires no memories. This design has 1596 input and output pins and contains three independent clock domains. The resulting post-route, optimized design has about 350k standard cell instances, not counting decoupling capacitance and fill. Using a mixed corner flow with power optimization effort level set high provides a 1 ns clock period at the worst-case SS corner. Emphasizing low standby power, the AES design (Fig. 10(b)) uses 38% SLVT cells to meet a 520 ps clock period at the TT 25ºC corner. Power optimization inserts 24% SRAM V_{th} cells to reduce leakage power consumption. They are predominately used in empty paths, i.e., flip-flop to flip-flop with no

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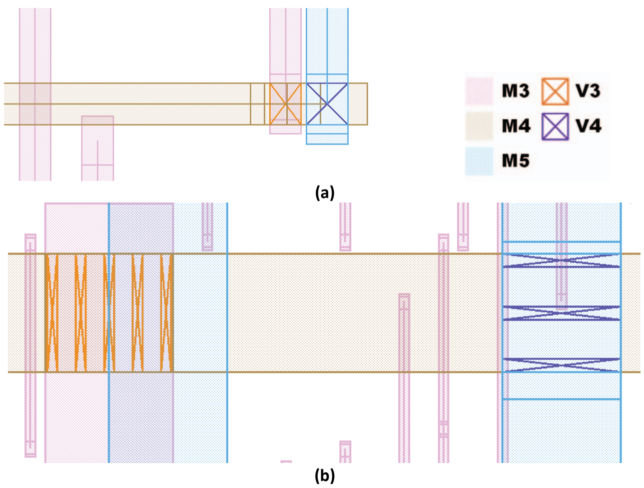

Fig. 11. Self-aligned vias in routing (a) and power (b). Non-square vias are required for signals on layers connecting to dissimilar width metals, e.g., M3-V3-M4, but square for same width, e.g., M4-V4-M5 (a). Power connections are consistent as SAV in the PDK (b). We force them on grid, following the DRCs that disallow off-grid edges. V2 are not shown.

intervening gate paths. APR designs with SRAMs developed on the PDK are under development, primarily to validate the SRAM liberty files and LEFs, rather than the cell libraries.

D. SAV in Routing and Power

The DRCs consistently enforce the use of SAV on all layers. Dissimilar routing layer width on adjacent layers forces rectangular, rather than square vias, as shown in Fig. 11(a). M3 (pink) differs from the M4 and M5 width (brown and blue, respectively). To be consistent in the power routing, wide vias, whose extents are defined by the wide power rail hard mask, are used as illustrated in Fig. 11(b) for the same layers. The technology LEF support for these has been an ongoing effort and requires specific APR flows that match. Specifically, the power rails must be constrained so the outer edges are coincident with those of signals on the outer tracks, as well as respecting the coloring. A track beginning on color A must end on color A, necessitating for example, upper power rail widths of 3, 5, 7, or 9 tracks.

E. Use in courses

The PDK and a number of cell library iterations have been used in graduate level VLSI design courses at Arizona State University multiple times over two years. The libraries have also been used in semi-custom control logic and decoders for SRAMs and register files. The cache EDAC of Fig. 10(a) above was one class project. This effort has exposed various small issues with the design rules and library blockage issues over time. As mentioned, the technology LEF remains a work in progress. Most of the work with the library has constrained M2 and M3 to 1-D routing, but custom designs have used M2/M3 2-D routing.

VI. Summary and Conclusions

This paper has presented 7.5-track cell libraries supporting multiple V_{th}'s on a 7 nm node finFET-based PDK. There are also other industry quality libraries under development for this PDK [16]. The resulting libraries are intended to allow realistic APR on an advanced process in an academic setting, i.e., research or coursework. Features to decrease cell size, to lower parasitics, limit leakage, and address reliability issues have been described. The libraries cleanly support SAQP compliant M2 and M3, although the required non-SAV V1 are not supported by the PDK—we believe wide M2 SAV may require library modification. The full APR collateral for routing and power distribution using the libraries has been completed for Cadence Innovus. The cells have been validated to ensure pin access and usability with small and large designs implemented in APR. Tricking Innovus to support sub-20 nm dimensions has been explained, and is part of the APR/library collateral. The scaled QRC technology file extraction correlates well with the Calibre extraction deck, that works on unscaled layout.

Acknowledgment

We gratefully acknowledge the contributions of Lovish Masand, Ankita Dosi, and Parv Sharma to the SRAM and APR setup efforts and experiments. We additionally thank ARM and specifically Saurabh Sinha, Brian Cline, Lu Shifren, and Greg Yeric, for their advice in the PDK development.

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